

Antiviral drugs

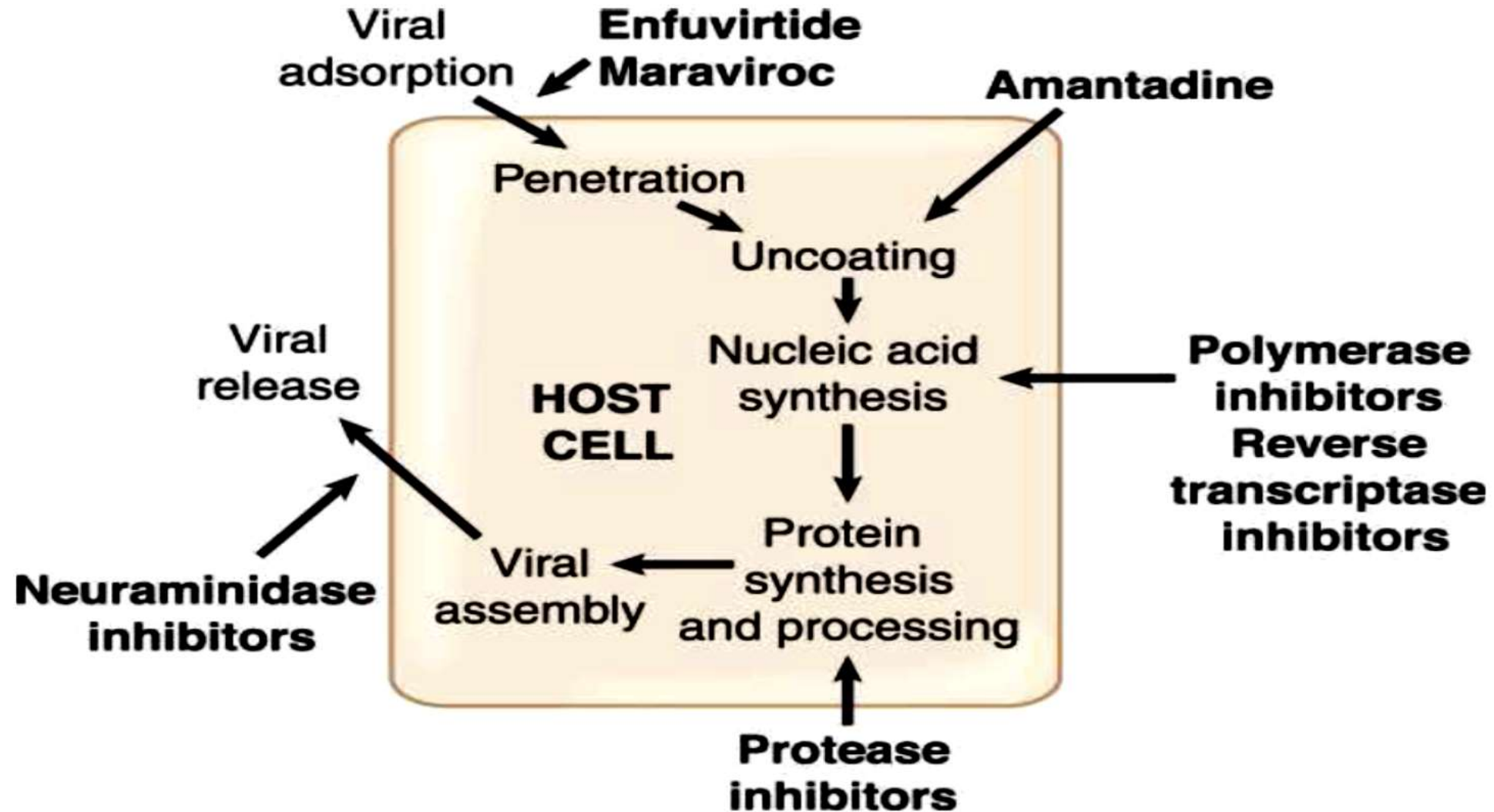
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Objectives

- Describe the mechanisms of action , pharmacokinetic properties and adverse effects of drugs used in respiratory infections e.g. influenza , herpes ,cytomegalovirus, Hepatitis and AIDS.“
- Apply the pharmacological effects to their major clinical applications.

Sites of action of Antivirals



- **I. Treatment of respiratory virus infections:**

- E.g. influenza A & B and respiratory syncytial virus (RSV).

- **A-Neuraminidase inhibitors:**

- **Mechanism of action:**

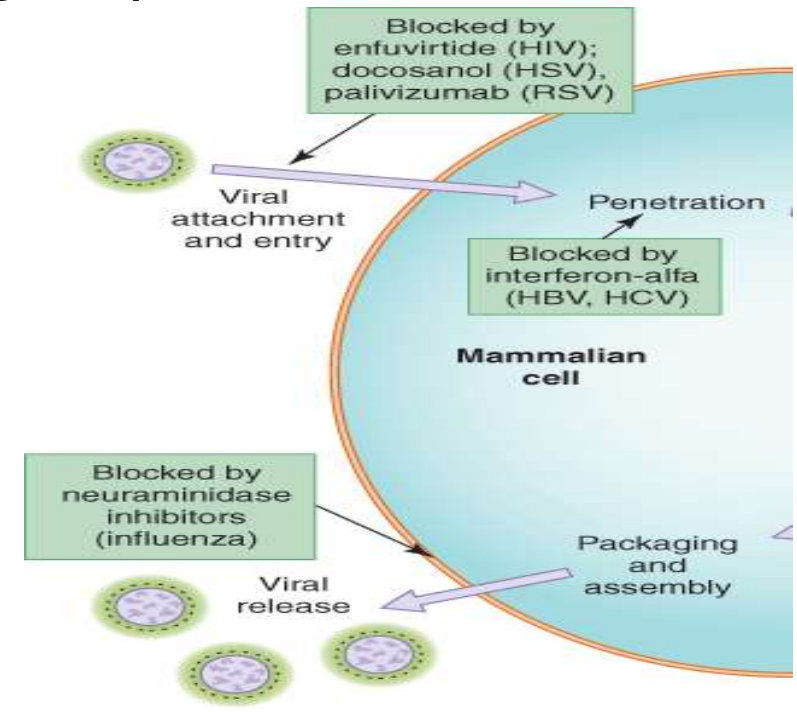
- Decrease duration and severity of the disease if administered in the *early 24-48 hours after infection* .

- E.g., **oseltamivir and zanamivir** (sialic acid analogues).

- They inhibit **neuraminidase enzyme, preventing the release of new virions.**

- Effective against **influenza A & B** (unlike amantadine).

- Don't interfere with immune response to influenza A vaccine.



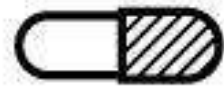
NDC 0004-0800-85

Tamiflu[®] (oseltamivir phosphate) Capsules

75 mg

Each capsule contains oseltamivir phosphate equivalent to 75 mg oseltamivir (free base).

R_x only



10 Capsules

Genentech

10175388

Zanamivir
Relenza



Pharmacokinetics:

- Oseltamivir is a prodrug which is active ***orally***.
- Zanamivir either ***inhaled or taken intranasal***.

Adverse effects:

- Oseltamivir → GIT discomfort.
- Zanamivir → irritation of respiratory tract (**avoided** in patients with severe asthma).

- **Resistance:**

- mutations of **neuraminidase** → however, mutant viruses are less infective & virulent than wild types.

B-Inhibitors of viral uncoating:

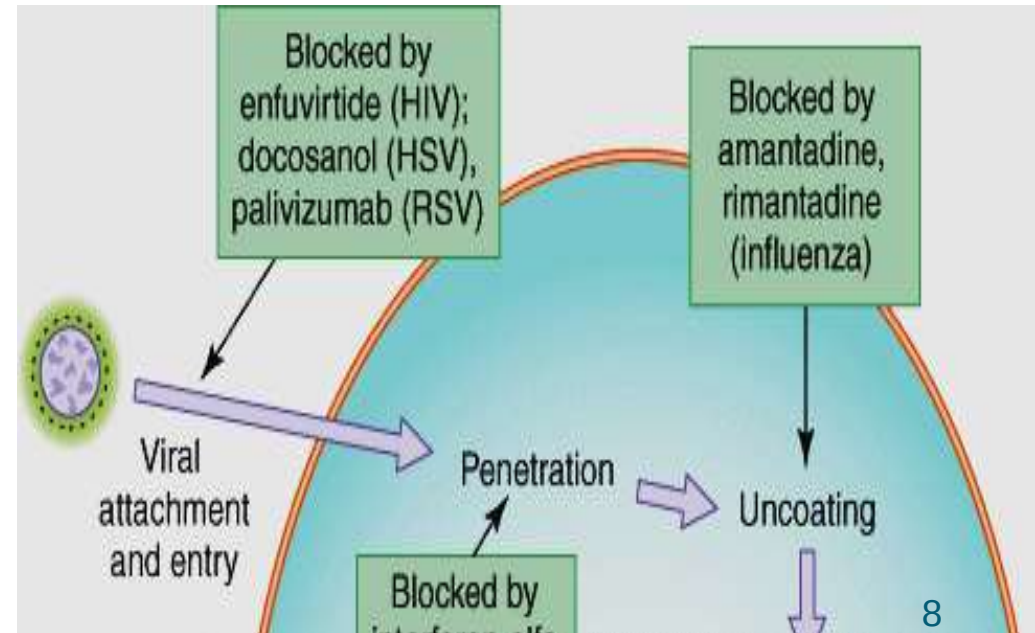
- e.g. **amantadine & rimantadine**.

- Neither drug impairs immune response to **influenza A**.
- Can be given as supplement to vaccination (until AB response occurs).
- Equally effective.



Mechanism of action:

- Inhibit **uncoating of viral RNA and release of new virions** into the host cell.



Pharmacokinetics:

Both are given orally.

- ***Amantadine crosses BBB*** and is excreted into urine.
- **Rimantadine** doesn't cross BBB and is extensively metabolized by the liver.
- It has a longer $t_{1/2}$.

- Adverse effects:

- Both cause GIT discomfort.
- ***Amantadine causes CNS effects*** (insomnia, slurred speech, ataxia & excitement (up to **convulsions**),.
- Use ***cautiously*** in patients with renal failure, epilepsy & psychiatric illness.
- Rimantadine has ***fewer CNS*** side effects (doesn't cross BBB).
- Both are **teratogenic** in rats so must be used with ***caution in pregnant and nursing women.***

- **C-Ribavirin:**
- **Mechanism of action:**
- Guanosine analogue triphosphated by host cell kinases inhibiting viral **mRNA capping & RNA-dependent RNA polymerase** → inhibit replication of wide range of RNA and DNA viruses (*rhinoviruses and enteroviruses are resistant*).
- **Pharmacokinetics:**
- Taken orally, IV and as aerosol which may be safer(in RSV infection).
- Doesn't cross BBB, excreted in urine.
- Monitoring is essential.

- Clinical uses:

- Broad spectrum against *DNA & RNA viruses*.
- *RSV* (not in adults), *viral hepatitis C, influenza A & B & Lassa virus* (of choice if started early).

- Adverse effects:

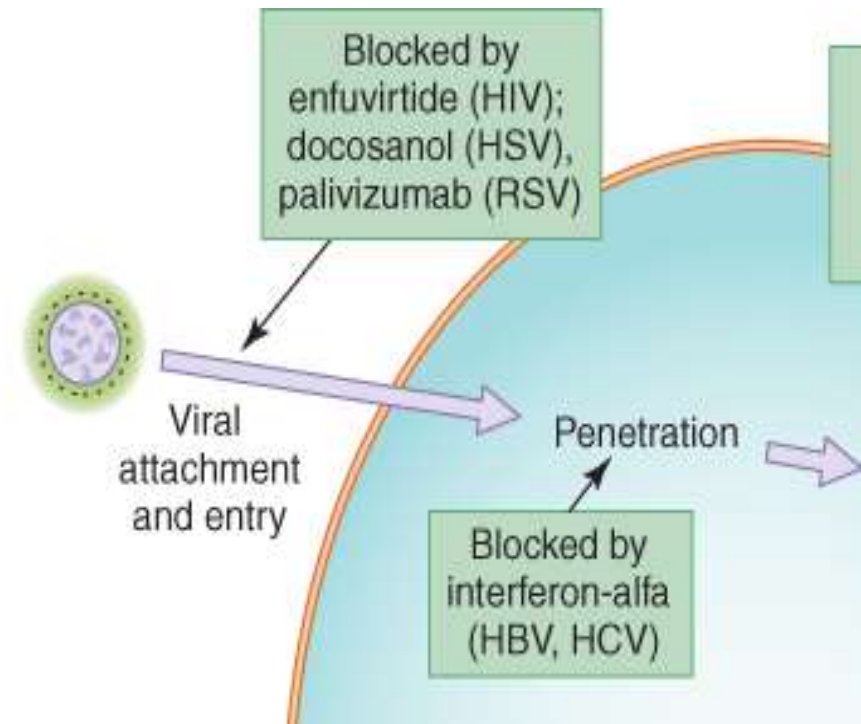
- Mild GIT disturbances (**orally**).
- Respiratory irritation, rash, conjunctivitis (**aerosol**).
- Teratogenicity and mutagenicity(*contraindicated in pregnancy*).
- Dose-dependent transient *hemolytic anaemia* (**10%**) (**extensively accumulate in RBC**).

II. Anti-hepatitis agents:

- -(A, B, C, D & E viruses)
- -Combination of interferon- β and lamivudine is no more effective than **lamivudine alone in hepatitis B.**
- -Patients with **hepatitis C** respond to combination of **interferon- β and ribavirin.**
- **A-Interferon:**
- **Antiviral, immunosuppressive and antiproliferative.
- Used in hepatitis *C and genital warts*

Mechanism of action of Interferon:

- -Induction of host cell enzymes that ***inhibit viral RNA translation into viral proteins & stops replication of the virus.***
- Pharmacokinetics:
- ***Not active orally***, given IV, SC or intra lesionally.
- Metabolized by liver.
- *Once a week dosing can be used.*



- Adverse effects:

- Chills, fever, **myalgia, arthralgia.**
- ↑ transaminases, nephrotoxic and neurotoxic.
- Dose-dependent **bone marrow depression** & potentiates myelosuppression of other drugs such as zidovuridine and azathioprine.
- GIT disturbances.
- Hair & weight loss and **sever fatigue.**
- ↑level of theophylline.
- Acute hypersensitivity reactions& hepatic failure are rare.
- Autoimmune thyroiditis, congestive heart failure.

B-Lamivudine:

- Inhibit both hepatitis B virus (HBV) DNA polymerase and HIV reverse transcriptase.
- Active orally, well distributed & renally excreted(70%).
- Dose adjustment **in renal** insufficiency.
- Chronic treatment improves biochemical markers and reduced hepatic inflammation.



C-Adefovir:

- **Discontinuation** →exacerbation of ***severe hepatitis (25%)***.
- Caution in patients with renal insufficiency.
- Should be used **cautiously** in patients with renal dysfunction.

D-Entecavir:

Discontinuation →**severe hepatitis**

Adefovir and Entecavir **can be used also for hepatitis B.**

E-Ribavarin

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III.Treatment of herpes virus infection:

- Herpesviruses are associated with **cold sore, viral encephalitis, and genital infections.**
- Drugs are only active during the acute phase of viral infection.
- All are purine or pyrimidine analogues that ***inhibit DNA synthesis except foscarnet and fomivirsen.***
- **A-Acyclovir (zovirax):**
- **Mechanism of action:**

Prodrug phosphorylated by host cells' kinases into active triphosphate →inhibits **viral DNA polymerase & terminates DNA chain elongation.**



- Pharmacokinetics:

- Taken IV, orally or topically.
- Well distributed (cross BBB).
- Partially metabolized, excreted by kidney.

- Therapeutic uses:

- Varicella-zoster infections (chicken pox & H. zoster).
- H. simplex type 1 & 2 (mucocutaneous, genital and **HSV encephalitis in which it is the drug of choice**).



- Adverse effects of acyclovir:

- GIT disturbances (**orally**).
- IV use → local inflammation if there is extravasations or renal dysfunction may occur (high doses or dehydrated patients) due to **crystalluria** (**adequate hydration + slow infusion rate**).
- Topically on the eye → stinging sensation.
- Headache & **CNS** effects (confusion, lethargy & tremors).
- Resistance.

B-Valacyclovir:

- Rapidly **converted to acyclovir** after oral administration → 3-5 **greater serum** levels than those with acyclovir(***better bioavailability***).
- Similar or slightly more efficacy than acyclovir.
- **Adverse effects:**
 - GIT manifestations, thrombocytopenic purpura.
 - In transplants patients, non- dose limiting effects.

- Treatment of cytomegalovirus (CMV) infection
- Ganciclovir:
 - Prodrug, activated to triphosphate which competitively inhibits DNA polymerase & causes termination of DNA elongation.
 - Drug of choice in *the life threatening CMV pneumonia (IV & orally) or retinitis* (intraocular implants) in immunocompromised patients and in prophylaxis against CMV in transplant patients.
 - IV administered, well distributed (crosses BBB), excreted into urine,
 - accumulated in renal failure.

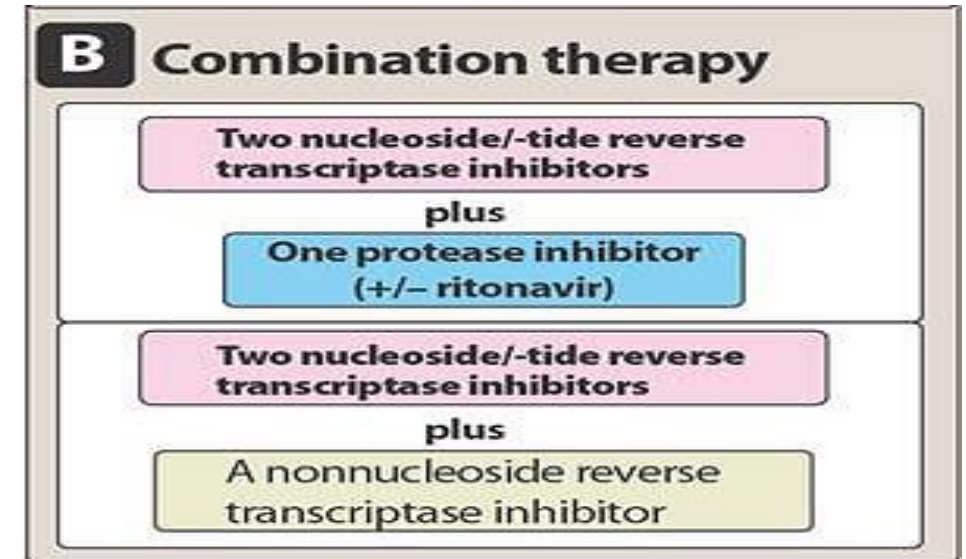


- Adverse effects:

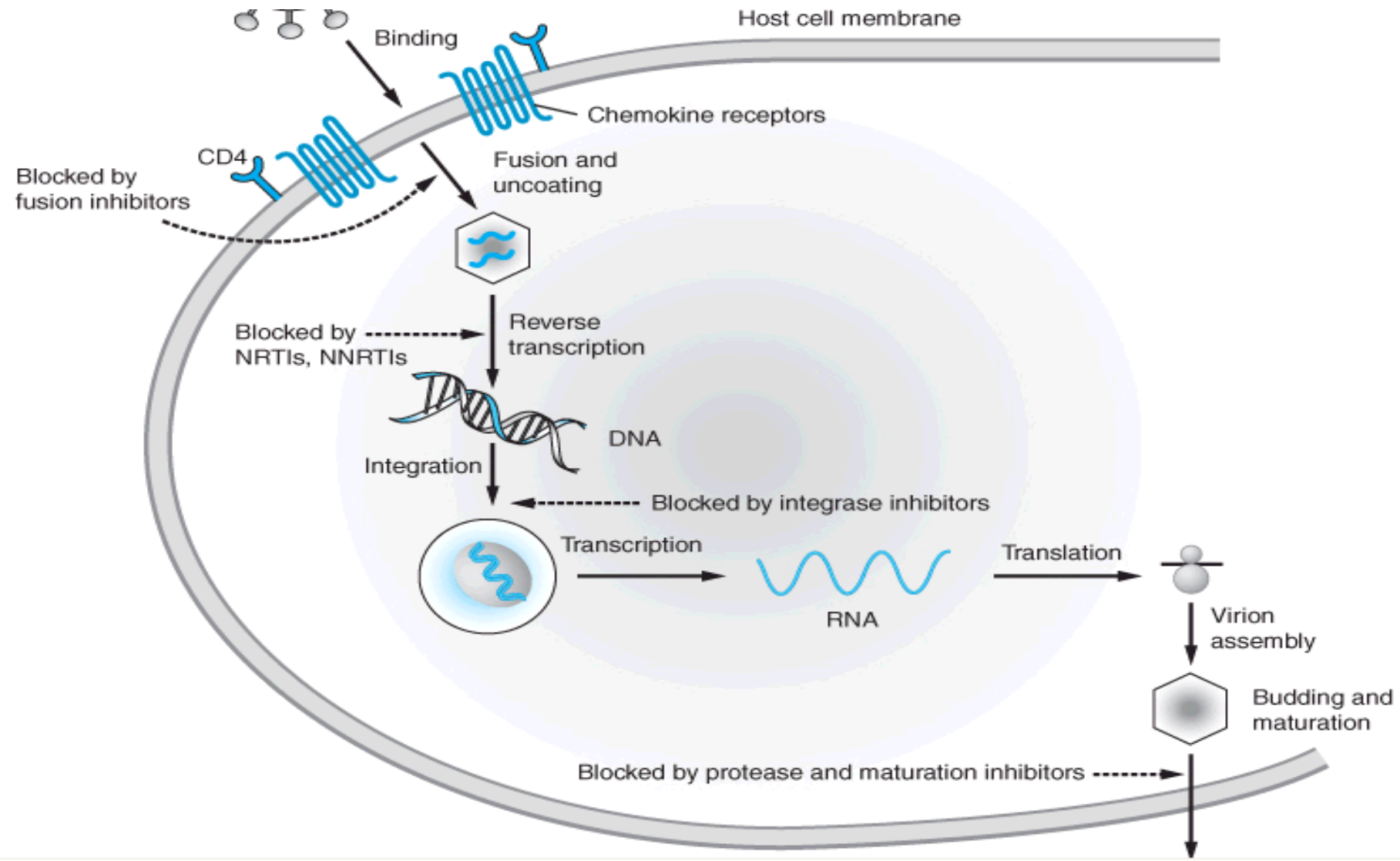
- **Myelosuppression** is the most common. Neutropenia is increased when combined **with zidovudine**, azathioprine.
- Potential carcinogenicity & teratogenicity.
- Vitreous hemorrhage & retinal detachment (with intraocular implants).

• Treatment of AIDS

- The most effective combinations (so called highly active anti- retroviral therapy *or HAART*).
- This comprises **two nucleoside reverse transcriptase inhibitors (NRTIS) with either** a non-nucleoside reverse transcriptase inhibitor (NNRTI) or a protease inhibitor (PI) (*avoid using drugs acting by the same mechanism or having same toxicity, avoid drug interactions, consider patient factors as concurrent diseases*). .
- This combination produces profound **suppression of viral replication** and results in useful **restoration of immune system**.
- There are **five classes of drugs**: NRTIS, NNRTI, protease inhibitor (PI), entry inhibitors and the integrase inhibitors



Life cycle of HIV. Binding of viral glycoproteins to host cell CD4 and chemokine receptors precedes **fusion** and entry into the cell. After uncoating, **reverse transcription** copies the single-stranded HIV RNA genome into double-stranded DNA, which is **integrated** into the host cell genome. Gene **transcription** by host cell enzymes produces messenger RNA, which is **translated** into proteins that assemble into **immature noninfectious virions** that bud from the host cell membrane. Maturation into **fully infectious virions** is through proteolytic cleavage



- **1-Nucleoside reverse transcriptase inhibitors (NRTIs):**

- -All are phosphorylated by host enzymes to triphosphate analogue which act by competitive inhibition of **HIV-1 reverse transcriptase** and can be incorporated into the growing viral DNA chain to cause its termination.
- -All (except lamivudine & abacavir) cause **fatal hepatic toxicity**(lactic acidosis and hepatomegally).

- **Zidovudine:**

- Same mechanism of action.
- Orally administered, well distributed including BBB.
- Most metabolized (glucoronylation) by liver, then excreted in urine.
- Used in **pregnancy to prevent prenatal infection.**

- Adverse effects:

- Mainly **myelosuppression** with long term use (neutropenia, thrombocytopenia).
- GIT upset, headache, flu-like syndrome.
- Megaloblastic anemia, myopathy.
- **Neurotoxicity** up to convulsions (very high doses).
- **Toxicity increases** with concomitant administration of drugs metabolized by glucoronylation as probenecid, acetaminophen, indomethacin, cimetidine...
- Ribavirin and stavudin should be **avoided** with zidovudine.

- Didanosine:
- Same mechanism of action.
- Adverse effects:
- Dose-dependent pancreatitis & peripheral neuropathy.
- Hepatitis, CNS toxicity, asymptomatic hyperurecemia.
- Should be taken 2 hours after drugs that **need acidity for absorption** (e.g. ketoconazole).
- Lamivudine: mentioned before.

• 2 -Non-nucleoside reverse transcriptase inhibitors (NNRTIs):

- They bind directly to a site (near that of NRTIs) on HIV-1 reverse transcriptase, resulting in blockade of **RNA-dependent DNA polymerase** activities.
- Unlike NRTIs, the NNRTIs neither compete with nucleoside triphosphatase nor require phosphorylation to be active.
- There is no cross-resistance between members of this class or between members of this class and NRTIs or PIs.
- Resistance develops rapidly with monotherapy.

- **Nevirapine:**

- -Effective in prevention of transmission of HIV from mother to newborn when administered to a mother at the onset of labor followed by a dose given to the neonate for 3 days (**as substitute of ziduvudine**).

- **Adverse effects:**

- Severe and **life threatening dermatologic** effects including Stevens-Johnson syndrome and toxic epidermal necrolysis. Gradual dose increase over 14 days is recommended to decrease serious epidermal reactions.
- **Fatal hepatotoxicity** (liver function tests must be serially monitored).
- **Efavirenz:** **Mainly CNS adverse effects**

- **3- Protease inhibitors(PIs e.g. naviars):**

- During replication of HIV, the mRNA is translated into *inert polyproteins which require a virus - specific **protease (HIV aspartyl protease)** to cleave them into structural and functional proteins of the mature virus*

-PIs disrupt this essential process.

CONT.,

- **Ritonavir** is no longer used as a single protease inhibitor but instead is used **as a pharmacokinetic enhancer or “booster”** of other protease inhibitors.
- The resulting higher levels of the “boosted” protease inhibitors also ***help to prevent the development of resistance.***
- Ritonavir is a potent inhibitor of CYP3A, and concomitant ritonavir administration (at low doses) increases the bioavailability of the second protease inhibitor, ***often allowing for longer dosing intervals.***
- **Cobicistat** is an inhibitor of cytochrome P450 (CYP) 3A enzymes that is indicated as a pharmacokinetic enhancer (i.e. booster) of the HIV-1 protease inhibitors (PIs) as **darunavir** in adults.

Adverse effects:

- All commonly cause GIT disturbances
- ↑TG & LDL levels & glucose intolerance.
- They are enzyme inhibitors & affected by enzyme inducers & inhibitors.
- Redistribution and **accumulation of body fat** with cushingoid appearance.
- **Serious drug interactions**, thus midazolam, warfarin, rifampicin, barbiturates & ergot alkaloids are to be avoided.
- Indinavir:
 - -In addition to adverse effects common to PIs, they can cause kidney stones; **avoided by adequate hydration.**



- 4- Viral fusion inhibitors(Entry inhibitors)

- Enfuvirtide:

- It is a fusion inhibitor that blocks entry into the host cell.
- It is given SC in combination with other anti retroviral agents.
- The most common **adverse effects** are local injection site reactions, hypersensitivity may occur, also eosinophilia but no drug interactions (no involvement of cytochrome p450 system).

5- Integrase inhibitors:

- Raltegravir:
- Inhibits the final step of integration of strand transfer of the viral DNA into our host cell DNA.
- **NB :**
- Entravirin:
- **2nd generation NNRTI** which is active against many resistant strains to the 1st generation.

Mechanisms of Action of Antiviral-Retroviral Drugs

Mechanism of Action	Major Drugs
Block viral penetration/uncoating	AmAntAdine, rimAntAdine
Inhibit viral neuraminidase	Zanamivir, oseltamivir
Inhibit viral DNA polymerases	Acyclovir , foscarnet, ganciclovir
Inhibit viral RNA polymerases	Foscarnet, ribavirin
Inhibit viral reverse transcriptase	Zidovudine, didanosine, zalcitabine, lamivudine, stavudine, nevirapine
Inhibit viral aspartate protease	Indinavir, ritonavir, saquinavir, nelfinavir